8 Figure 8.1 shows the variation with nucleon number A of the binding energy per nucleon B_E of a nucleus.

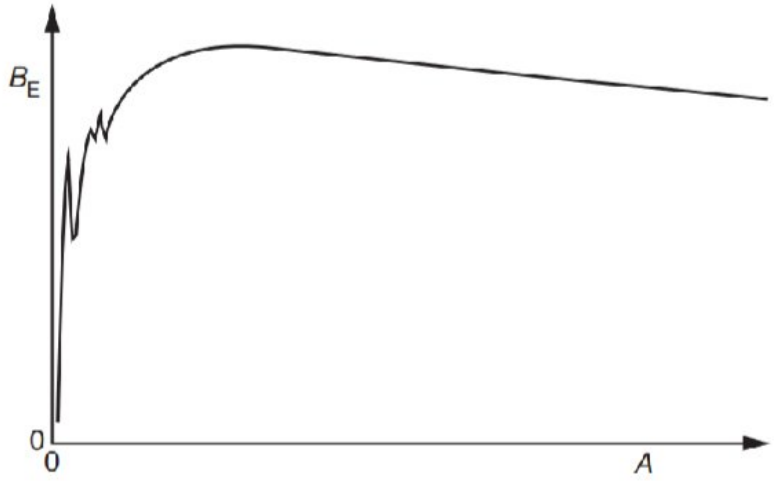


Figure 8.1

When a nucleus of uranium-235 (${}^{235}_{92}\mathrm{U}$) absorbs a neutron in a nuclear reactor, it undergoes fission to form various products and release further neutrons.

In one nuclear reaction involving the fission of uranium-235, barium-141 (${}^{141}_{56}\mathrm{Ba}$) and krypton-92 (${}^{92}_{36}\mathrm{Kr}$) are the products. Three neutrons are also released when these products are formed, and energy is released.

The equation for this reaction is shown:

$$ {}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} $$

(a) Explain, with reference to the line in Figure 8.1, why this reaction results in the release of energy.

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..... [2]

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(b) Uranium-235 has a binding energy per nucleon of 7.5909 MeV.

Data for some masses are given in Table 8.1

	mass/u
proton	1.00814
neutron	1.00898
barium-141	140.91441
krypton-92	91.92615

Table 8.1

(i) Show, to 5 decimal places, that the mass of a uranium-235 nucleus is 235.12259u.

[2]

(ii) Calculate the energy released in this fission reaction.

energy = ... J [2]

(iii) The total kinetic energy of the products for the reaction is found to be more than the answer in **(b)(ii)**.

Suggest a reason for this difference.

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..... [1]

[Total: 7]

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